Outline

1. Indefinite integral
2. Definition of the definite integral
3. Properties of the definite integral
4. Fundamental Theorem of Calculus
5. Methods of Substitution and Integration by Parts
6. Volumes
7. Arc Length and Surface Area

• Substitution rule for definite integrals

If $g'(x)$ is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

There are two ways we can make a substitution: either $u = g(x)$ or $x = h(u)$.

Example

Find $\int_0^2 \sqrt{x}\, dx$.

Solution. We can do it directly:

$$\int_0^2 \sqrt{x}\, dx = \left[\frac{2}{3}x^{\frac{3}{2}}\right]_0^2 = \frac{2}{3}2^{\frac{3}{2}} = \frac{2\sqrt{8}}{3}.$$

Alternatively, we can use the substitution

$$u = \sqrt{x} \geq 0 \implies du = \frac{1}{2\sqrt{x}}\, dx \quad \text{or} \quad dx = 2u\, du.$$

As x ranges from 0 to 2, we have u runs from 0 to $\sqrt{2}$. Hence, we have

$$\begin{aligned}\int_0^2 \sqrt{x}\, dx &= \int_0^{\sqrt{2}} 2u^2\, du \\ &= \left[\frac{2u^3}{3}\right]_0^{\sqrt{2}} = \frac{2\sqrt{8}}{3}.\end{aligned}$$

• Integrals of symmetric functions

From the Substitution rule, we can get the following results.

• Integrals for even and odd functions

Suppose f is continuous on the closed interval $[-a, a]$.

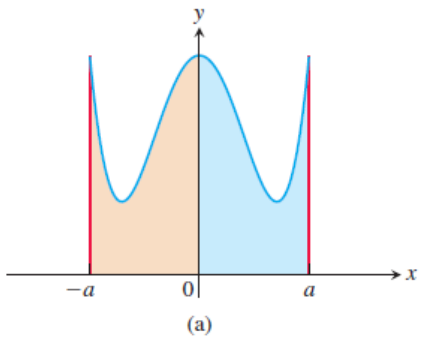
- If f is even on $[-a, a]$ (i.e., $f(-x) = f(x)$, for all $x \in [-a, a]$), then

$$\int_{-a}^{a} f(x)\, dx = 2 \int_{0}^{a} f(x)\, dx.$$

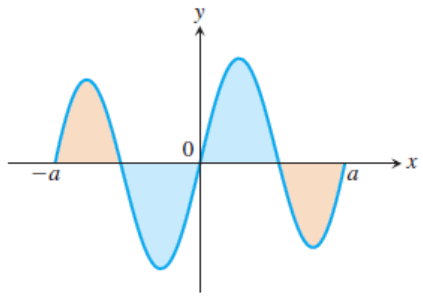
- If f is odd on $[-a, a]$ (i.e., $f(-x) = -f(x)$, for all $x \in [-a, a]$), then

$$\int_{-a}^{a} f(x)\, dx = 0.$$

The illustrations for the theorem above:



(a)



(b)

• Integration by Parts

• Integration by Parts for definite integrals

If f and g are both differentiable, then

$$\int_a^b f(x)g'(x)\,dx = \Big[f(x)g(x)\Big]_a^b - \int_a^b f'(x)g(x)\,dx.$$

Since the formula is also often written in the compact form

$$\int u\,dv = uv - \int v\,du,$$

the Integration by Parts formula does not always work, because you are basically transferring the task of evaluating $\int u\,dv$ to the task of evaluating $\int v\,du$.

Example

Find $\int_1^2 \ln x \, dx$.

Solution. We choose $\begin{cases} u = \ln x \\ dv = 1 \, dx. \end{cases} \implies \begin{cases} du = \frac{1}{x} \, dx \\ v = x \end{cases}$ and apply the IBP formula

$$\begin{aligned} \int_1^2 \ln x \, dx &= \left[uv\right]_1^2 - \int_1^2 v \, du = \left[x \ln x\right]_1^2 - \int_1^2 x \cdot \frac{1}{x} \, dx \\ &= (2 \ln 2 - 1 \ln 1) - \int_1^2 1 \, dx = 2 \ln 2 - \left[x\right]_1^2 \\ &= 2 \ln 2 - 1. \end{aligned}$$

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Summary

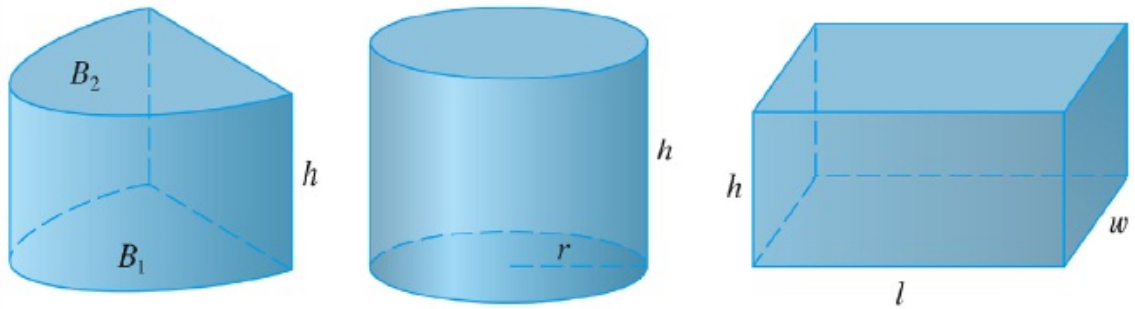
In this section, our study moves to a 3-dimensional setting (volume) versus the previous 2-dimensional setting (area).

We use the definite integral to help us to make precise what the volume of a solid means.

So, what is the “volume" of a given solid?

Again this is easy to answer for “regular" solids for which we have standard formulas:

- The volumes of cylindrical solids.



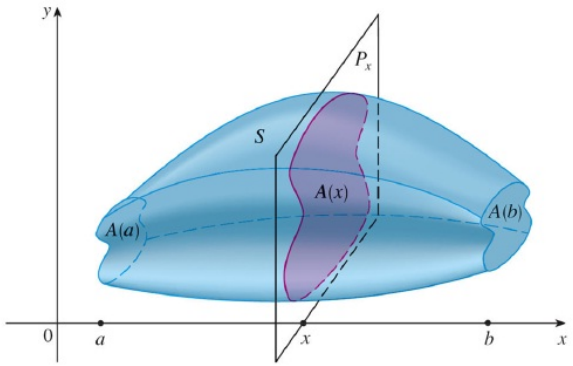
(a) Cylinder
$V = Ah$

(b) Circular cylinder
$V = \pi r^2 h$

(c) Rectangular box
$V = lwh$

However the volume is not so easy to find for a non-cylindrical case.

- A solid S with a cross-section highlighted.



To help us analyze this problem, we consider intersecting the given solid with a plane P_x. This plane P_x is perpendicular to the x-axis and passes through the point x on the x-axis, where $a \leq x \leq b$.

The common region between P_x and S is called the cross-section of S, and the area of the cross section is denoted by $A(x)$.

In general, we have

$$V = \int_a^b A(x)\, dx.$$

How to derive this formula? There are two main ways of calculating volume of a given solid:

- Volumes using cross-sections.
- Volumes using cylindrical shells.

• Volumes using cross-sections

We study how to calculate the volume of a given solid using the cross-sectional method.

There are three types of solids where the cross-sectional method applies:

1. General solid.
2. Solid of revolution (disk method).
3. Solid of revolution (washer method).

- **Cross-sectional method I: General solids**

This method can be applied to any given solid, as long as the cross-sectional area $A(x)$ can be easily found.

Let's consider the solid. We apply the same idea as we previously did for the area problem - **see Appendix !**

So we have the following definition.

Definition. Let S be a solid that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane P_x through x and perpendicular to the x-axis is given by the continuous function $A(x)$, then the volume of the solid S is

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*) \cdot \Delta x = \int_a^b A(x)\, dx.$$

Some things to remember.

1. It is important to keep in mind that when we use the volume formula $V = \int_a^b A(x)\,dx$, the area $A(x)$ is the area of the cross-section obtained by slicing through the point x perpendicular to the x-axis. This is usually not constant and is a function of x.

2. The variable x here refers to the x-coordinate. It can be negative, but must lie between a and b.

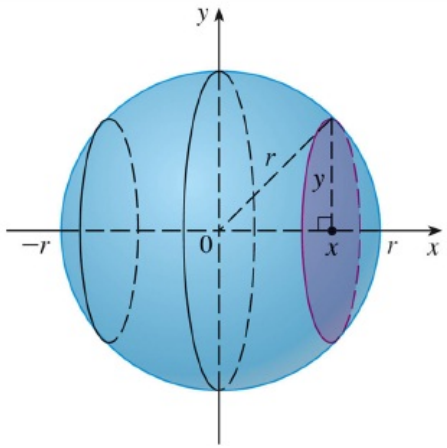
3. $A(x)$ must be positive.

Example

Show that the volume of a sphere of radius r is

$$V = \frac{4}{3}\pi r^3.$$

Solution. We place the sphere so that its center is at the origin. So we have a sphere of radius r, with a cross-section at x.



The cross section at x is a circle with radius $y = \sqrt{r^2 - x^2}$. Therefore, the cross-sectional area at x is

$$A(x) = \pi \cdot y^2 = \pi \cdot (r^2 - x^2).$$

Note that x refers to the x-coordinate value and may take on negative values. However the expressions for y and $A(x)$ are always positive. So we can compute the volume of the sphere using the formula

$$\begin{aligned}
\int_{-r}^{r} A(x)\,dx &= \int_{-r}^{r} \pi(r^2 - x^2)\,dx = \pi r^2 \int_{-r}^{r} dx - \pi \int_{-r}^{r} x^2\,dx \\
&\underbrace{=}_{\text{(by the FTC-2)}} \pi r^2 \Big[x\Big]_{-r}^{r} - \pi \left[\frac{x^3}{3}\right]_{-r}^{r} \\
&= \pi r^2(2r) - \frac{\pi}{3}\left(2r^3\right) = \frac{4}{3}\pi r^3.
\end{aligned}$$

- **Cross-sectional method II: Solid of revolution (disk method)**

This method can be applied to a second kind of solid.

The solid generated by rotating (or revolving) a plane region about an axis in its plane is called a solid of revolution.

The following example demonstrates how to find the volume of such a solid.

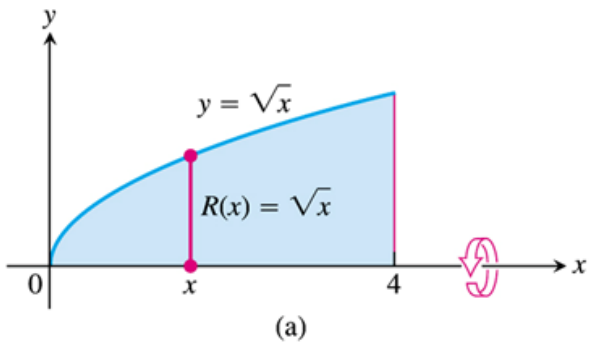
Example

Find the volume of the solid obtained by rotating about the x-axis the region under the curve $y = \sqrt{x}$ from 0 to 4.

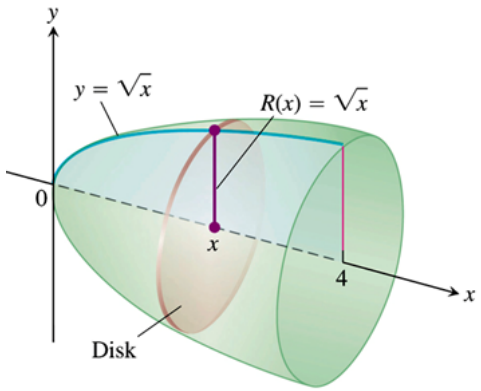
Solution. Let us try and figure out what the solid described in the question looks like.

- First, we start with the curve $y = \sqrt{x}$ in the 2-dimensional xy-plane (the region (shaded) under the curve $y = \sqrt{x}$ from $x = 0$ to $x = 4$).

- Next, we take this shaded region and rotate it 360° around the x-axis to get the 3-dimensional solid.



(a)



(b)

Now what do we get when we slice the plane P_x perpendicular to the x-axis through this solid to get the cross-section?

The area of the resulting cross-section should be a circle of radius $R(x) = \sqrt{x}$. So,

$$A(x) = \pi[R(x)]^2 = \pi\left(\sqrt{x}\right)^2 = \pi x.$$

Thus, since the solid lies between $x = 0$ to $x = 4$, the volume is

$$V = \int_0^4 A(x)\, dx = \int_0^4 \pi x\, dx = \pi\left[\frac{x^2}{2}\right]_0^4 = 8\pi.$$

We can also find the volume of the solid obtained by rotating about the y-axis the region bounded.

In this case, to take the cross section, we slice a plane P_y at the point with y-coordinate y, perpendicular to the y-axis, and hence, we integrate with respect to y.

In summary for the **disk method**, we have

- Volume by Disks Rotation about the x-axis:

$$V = \int_a^b A(x)\,dx = \int_a^b \pi[R(x)]^2\,dx.$$

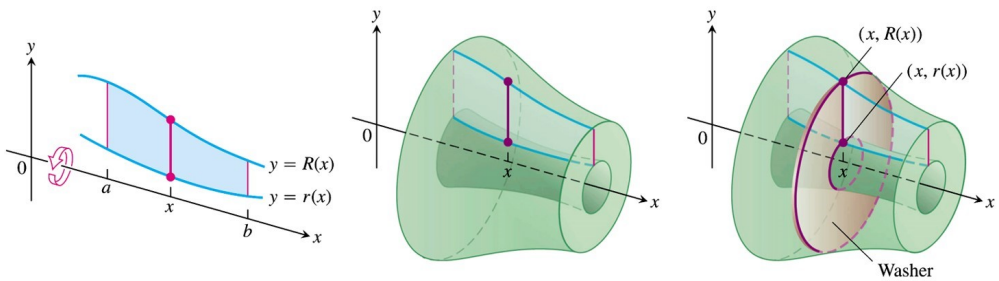
- Volume by Disks Rotation about the y-axis:

$$V = \int_c^d A(y)\,dy = \int_c^d \pi[R(y)]^2\,dy.$$

- **Cross-sectional method III: Solid of revolution (washer method)**

We now talk about the third kind of solid you may encounter in the cross-sectional method.

This kind of solid is usually obtained by rotating around the x- or y-axis a region bounded between two or more curves. The resulting cross-sectional area often looks like the region between two concentric circles and hence this is called the "washer" method.

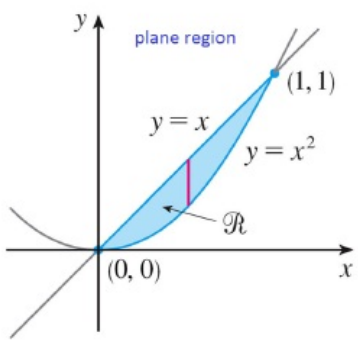


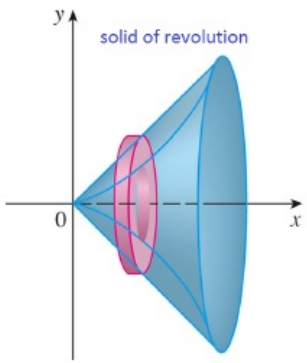
Example

Find the volume of the solid obtained by rotating about the x-axis the region enclosed by $y = x$ and $y = x^2$.

Solution. - First we sketch the 2-dimensional figure described in the question, and sketch the 3-dimensional solid obtained by rotating the 2-dimensional region about the x-axis.

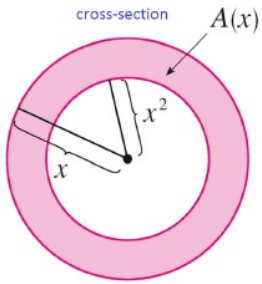
The region enclosed by $y = x$ and $y = x^2$, and the solid of revolution are below:





- Since the solid is obtained by revolving around the x-axis, we integrate with respect to x. To obtain the cross-section we must slice perpendicular to the x-axis. It has the shape of a washer with inner radius x^2 and outer radius x.

The cross-sectional area.



The cross-sectional area is thus

$$A(x) = \pi x^2 - \pi(x^2)^2 = \pi(x^2 - x^4).$$

Since the solid lies between $x = 0$ and $x = 1$, so the volume of the solid of revolution is

$$V = \int_0^1 A(x)\, dx = \int_0^1 \pi(x^2 - x^4)\, dx = \pi\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \frac{2\pi}{15}.$$

Thus, for the **washer method**, we have.

- Volume by Washers for Rotation about the x-axis:

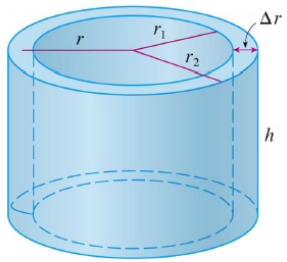
$$V = \int_a^b A(x)\, dx = \int_a^b \pi\left([R(x)]^2 - [r(x)]^2\right)\, dx.$$

• Volumes using cylindrical shells

After this study you should be able to:

1. Calculate the volume of a given solid using the cylindrical shell method.
2. Determine which method to use (cross-section versus cylindrical shells).

Consider the following cylindrical shell. It has inner radius r_1, outer radius r_2 and height h.



The volume of such a cylinder is given by

$$\begin{aligned} V &= \pi(r_2)^2 h - \pi(r_1)^2 h = \pi(r_2^2 - r_1^2)h \\ &= \pi(r_2 + r_1)(r_2 - r_1)h \\ &= 2\pi \cdot \frac{r_2 + r_1}{2} \cdot h \cdot (r_2 - r_1). \end{aligned}$$

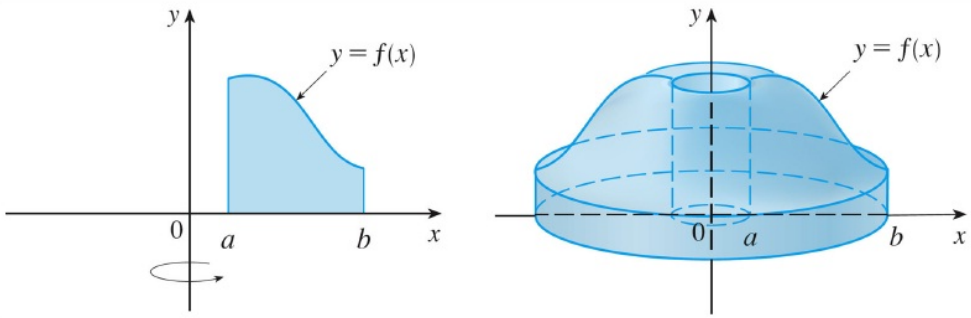
If we let $\Delta r = r_2 - r_1$ representing the thickness of the shell, and $r = \dfrac{r_2 + r_1}{2}$ representing the average radius of the shell, then the formula for the volume of a cylindrical shell becomes

$$V = 2\pi r \cdot h \cdot \Delta r.$$

This can be remembered as the product of $2\pi r$ (average circumference), h (height) and Δr (thickness).

A question: How to find the volume of such a solid of revolution?

The solid S is obtained by rotating about the y-axis the region under the curve $y = f(x)$ from $x = a$ to $x = b$, where $b > a \geq 0$:



We can use the previous method of taking cross-section to find the volume of the solid S.

How would we do it?

There is an alternative method to find volumes using cylindrical shells.

Namely, the same method as before is applied - **see Appendix !**

The **formula** to memorize: is as follows.

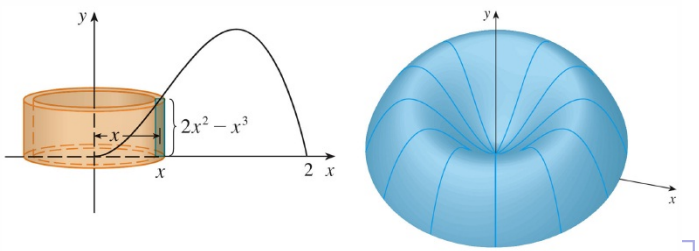
- Volume of cylindrical shells:

$$V = 2\pi r \cdot h \cdot \Delta r = \text{Average circumference} \times (\text{height}) \times (\text{thickness}).$$

Example

Find the volume of the solid obtained by rotating about the y-axis the region bounded by $y = 2x^2 - x^3$ and $y = 0$.

Solution. The region and the solid are sketched below:



As we can see from the diagram, a typical cylindrical shell at coordinate value x has radius x and height $f(x)$. So the volume of a typical approximating cylindrical shell is

$$(2\pi x) \cdot (2x^2 - x^3) \cdot \Delta x.$$

Then the volume of the solid is the definite integral of the above expression, i.e.

$$\begin{aligned} \int_0^2 (2\pi x) \cdot (2x^2 - x^3)\, dx &= 2\pi \int_0^2 (2x^3 - x^4)\, dx \\ &= 2\pi \left[\frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 = 2\pi \left(8 - \frac{32}{5} \right) = \frac{16\pi}{5}. \end{aligned}$$

Question: Can you try to solve the example using cross-sections?

So a natural question to ask is: Cross-section versus cylindrical shells, which one to use?

It really depends on how easy it is to find the cross-sectional area versus how easy to find the information on the cylindrical shells.

Often both methods are fine (as we have seen in the examples above).

However, a **general guideline** is as follows:

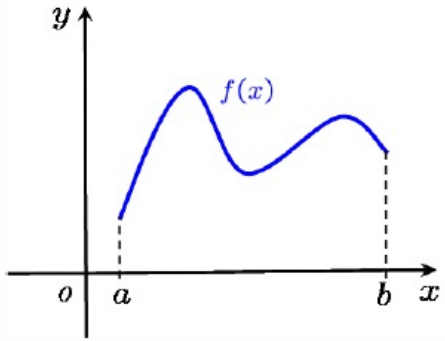
- Rotation of a given plane region about the x-axis:
  - If using cross-sectional method, slice perpendicular to the x-axis, and integrate w.r.t. x.
  - If using cylindrical shells, consider typical cylinders parallel to x-axis, and integrate w.r.t. y.
- Rotation of a given plane region about the y-axis:
  - If using cross-sectional method, slice perpendicular to the y-axis, and integrate w.r.t. y.
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We now discuss how to find the arc length of a smooth curve on the xy-plane.

For instance, a smooth curve defined by a differentiable function $f(x)$.

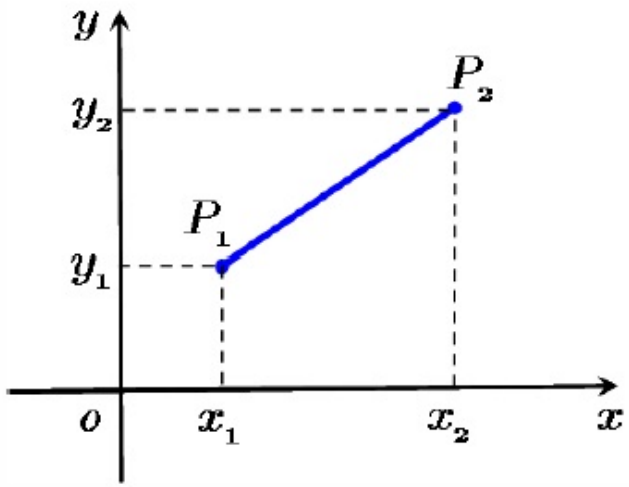


The basic idea is still the same as for finding volumes: divide, approximate, perform the sum, and take the limit.

- **Length of straight line segments**

Let's first figure out how to compute the arc length when we are given the simplest kind of smooth curve, a straight line segment.

Consider the line segment joining two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$.



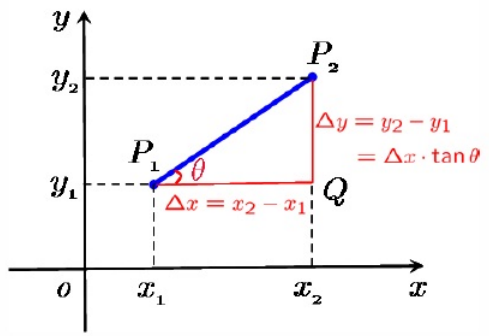
The length of this straight line segment is simply given by the distance between the two points P_1 and P_2:

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

The arc length can be found in another way by using the angle of elevation of the line segment - **see the Appendix !**

The arc length can be found in another way by using the angle of elevation of the line segment - **see the Appendix !**
We try to write this distance in another way.

We use θ as the angle of elevation of the line segment.



As shown in the figure, we may let $\Delta x = x_2 - x_1$ and $\Delta y = y_2 - y_1$.

So the distance between the two points can be written as:

$$|P_1P_2| = \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

Now consider the triangle in the figure. Since the angle of elevation of the line segment is θ, we have

$$\tan\theta = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \iff \Delta y = \Delta x \cdot \tan\theta,$$

and so substituting into the expression for $|P_1P_2|$ gives the formula for the length of a straight line segment

$$\begin{aligned} |P_1P_2| &= \sqrt{(\Delta x)^2 + (\Delta y)^2} \\ &= \sqrt{(\Delta x)^2 + (\Delta x \cdot \tan\theta)^2} = \sqrt{1 + \tan^2\theta} \cdot \Delta x. \end{aligned}$$

Note that $\tan\theta = \dfrac{\Delta y}{\Delta x}$ is the slope of the line passing through P_1, P_2.

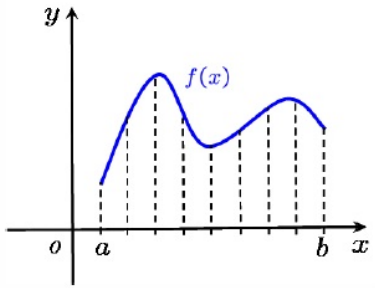
- **Deriving the formula for the length of an arc**

Now consider an arc lying on a curve which is not necessarily a straight line.

We use the same trick we have used many times before - **a self-study**.

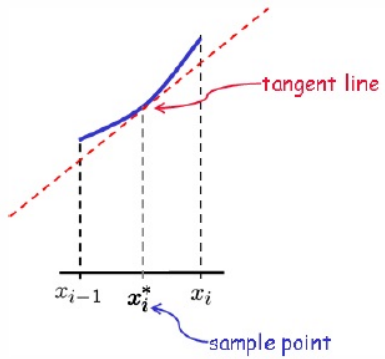
. . . . . .

- First divide the curved arc C (bounded by $x = a$ and $x = b$) into many small parts (tiny arcs) of equal width Δx (using a partition of $[a, b]$ by n subintervals $[a, x_1]$, $[x_1, x_2]$, $[x_2, x_3], \cdots$, $[x_{n-1}, b]$).



Each tiny arc on C looks roughly like a straight line segment.

- Next, to approximate this tiny arc by a straight line segment, we need to choose a straight line segment close to the tiny arc. But which line segment should we choose? It turns out that it is easiest to pick a sample point $x_i^* \in [x_{i-1}, x_i]$ *from the interval, and then taking the tangent line to the curve* $y = f(x)$ *at the point* $\left(x_i^*, f(x_i^*)\right)$.



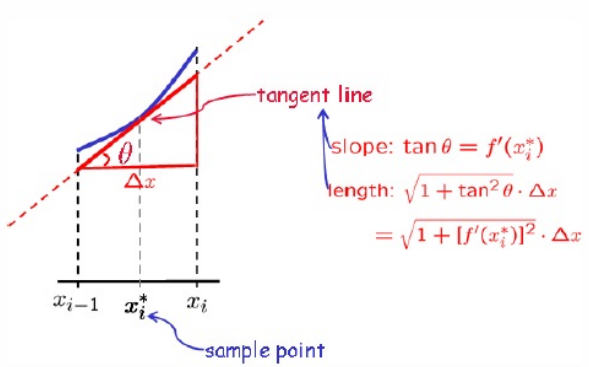
So we use the segment of the tangent line inside the interval $[x_{i-1}, x_i]$ *to approximate the tiny arc in the same interval* $[x_{i-1}, x_i]$.

To find the length of the approximating segment on the tangent line, we apply the formula we derived previously

$$\sqrt{1+\tan^2\theta}\cdot\Delta x.$$

We know the value of $\Delta x = x_i - x_{i-1}$. How do we find θ?

Let's put in the right angled triangle into the diagram



We note that the slope of this tangent is $f'(x_i^*)$, because the approximating line segment lies on the tangent line to the curve $y = f(x)$ at the point $\left(x_i^*, f(x_i^*)\right)$. This means that

$$\tan\theta = f'(x_i^*).$$

So we can apply the formula we obtained earlier for the length of a straight line segment:

$$\sqrt{1+\tan^2\theta}\cdot\Delta x = \sqrt{1+\left(f'(x_i^*)\right)^2}\cdot\Delta x.$$

This expression gives us the approximate arc length for each tiny arc.

- To find an approximate value for the arc length L of the entire curve C, we sum up the lengths to get:

$$L \approx \sum_{i=1}^{n} \sqrt{1+\left(f'(x_i^*)\right)^2}\cdot\Delta x.$$

- Finally, as usual, this approximation gets closer to L as $n \to \infty$, so to get L we take limit:

$$L = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{1 + \left(f'(x_i^*)\right)^2} \cdot \Delta x.$$

At this point, we should recognize the expression $\sum_{i=1}^{n} \sqrt{1 + \left(f'(x_i^*)\right)^2} \cdot \Delta x$ *as the* n^{th} *Riemann sum of the function* $\sqrt{1 + (f'(x))^2}$.

So by the definition of the definite integral, we get

$$L = \sum_{i=1}^{n} \sqrt{1 + \left(f'(x_i^*)\right)^2} \cdot \Delta x = \int_a^b \sqrt{1 + (f'(x))^2}\, dx.$$

This gives us the following **formula**:

Definition (Arc length formula)

If $f'(x)$ is continuous on $[a, b]$, then the arc length of the curve $y = f(x)$ from the point $A(a, f(a))$ to the point $B(b, f(b))$ is given by

$$L = \int_a^b \sqrt{1 + (f'(x))^2}\, dx.$$

You may choose to remember the formula in the form

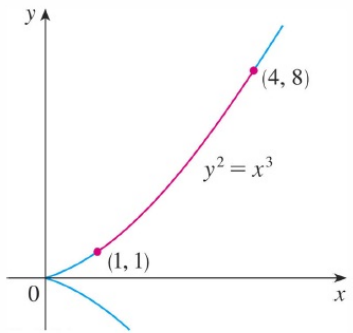
$$L = \int_a^b \sqrt{1 + (y')^2}\, dx.$$

Note that we require the function $f(x)$ to be not only differentiable, but to be continuously differentiable, i.e. $f'(x)$ is continuous. This ensures that the limit of the Riemann sum exists.

Example

Find the arc length of the semi-cubical parabola $y^2 = x^3$ between the points $(1, 1)$ and $(4, 8)$.

Solution. This is sketched in the figure below (the arc is colored in red).



We are only interested in the top half where $y \geq 0$ (since this is where the arc is defined). So the equation of the curve is in fact $y = x^{\frac{3}{2}}$.

To apply the formula for the arc length, we must first find the first derivative:

$$y' = \frac{3\sqrt{x}}{2}.$$

Now the arc length is given by the formula:

$$\begin{aligned}
\text{arc length} &= \int_1^4 \sqrt{1+(y')^2}\,dx = \int_1^4 \sqrt{1+\left(\frac{3\sqrt{x}}{2}\right)^2}\,dx \\
&= \int_1^4 \left(1+\frac{9}{4}x\right)^{\frac{1}{2}}\,dx = \left[\frac{\left(1+\frac{9}{4}x\right)^{\frac{3}{2}}}{\frac{3}{2}\cdot\frac{9}{4}}\right]_1^4 \\
&= \frac{8}{27}\left[\left(1+\frac{9}{4}x\right)^{\frac{3}{2}}\right]_1^4 = \frac{1}{27}\left(80\sqrt{10}-13\sqrt{13}\right).
\end{aligned}$$

- **Arc length function**

Some applications require a function that measures the arc length of a curve from a particular starting point to any other point on the curve.

Suppose that a smooth curve has the equation $y = f(x)$ for $a \leq x \leq b$. Let $s(x)$ be the length of the arc along the curve $y = f(x)$ from the point $(a, f(a))$ to the point $(x, f(x))$. Then we can view $s(x)$ as a function (what is the input and what is the output of $s(x)$?)

By the arc length formula, we have

$$s(x) = \int_a^x \sqrt{1 + (f'(t))^2}\, dt.$$

Example

Find the arc length function for the curve $y = x^2 - \dfrac{\ln x}{8}$ by taking $(1, 1)$ as the starting point.

Solution. First we compute the first derivative $y' = 2x - \dfrac{1}{8x}$. Then

$$\begin{aligned} s(x) &= \int_1^x \sqrt{1 + \left(2t - \frac{1}{8t}\right)^2}\, dt \\ &= \int_1^x \sqrt{4t^2 + \frac{1}{2} + \frac{1}{64t^2}}\, dt \\ &= \int_1^x \sqrt{(2t)^2 + \frac{1}{2} + \left(\frac{1}{8t}\right)^2}\, dt \\ &= \int_1^x \sqrt{(2t)^2 + 2(2t)\left(\frac{1}{8t}\right) + \left(\frac{1}{8t}\right)^2}\, dt. \end{aligned}$$

We note that the expression inside the square root as of the form

$$\underbrace{(2t)^2}_{p^2} + \underbrace{2(2t)\left(\frac{1}{8t}\right)}_{2pq} + \underbrace{\left(\frac{1}{8t}\right)^2}_{q^2} = \underbrace{\left(2t+\frac{1}{8t}\right)^2}_{(p+q)^2}.$$

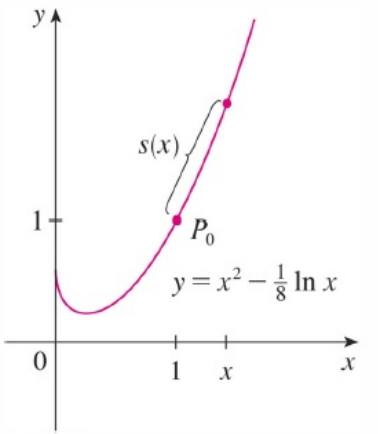
Then we continue

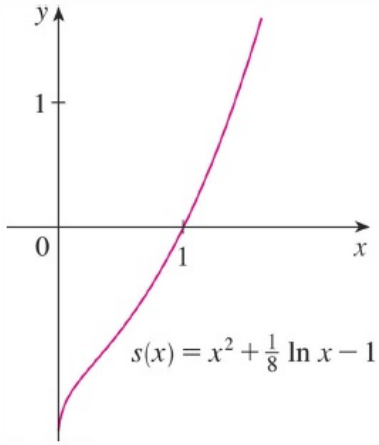
$$s(x) = \int_1^x \sqrt{\left(2t+\frac{1}{8t}\right)^2}\, dt = \int_1^x \left|2t+\frac{1}{8t}\right|\, dt.$$

Since t ranges from 1 to x, we have $2t + \frac{1}{8t} > 0$ and so

$$s(x) = \int_1^x 2t+\frac{1}{8t}\, dt = \left[t^2 + \frac{\ln t}{8}\right]_1^x = x^2 + \frac{\ln x}{8} - 1.$$

So the curve of the function $y = x^2 - \dfrac{\ln x}{8}$ and the curve of its arc length function $s(x) = x^2 + \dfrac{\ln x}{8} - 1$ are plotted in figures below.



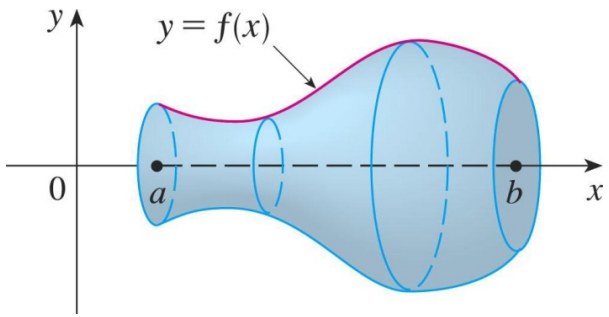


(Why do you think that $s(1) = 0$? Why is it that $s(x) < 0$ for $0 < x < 1$?)

- **Surface area**

We have already discussed the problem of finding the volume of a solid of revolution. How do we find the surface area of a solid of revolution? The key lies in using the arc length.

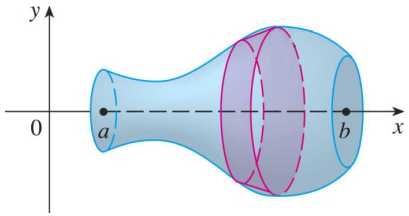
Consider the following solid of revolution given in the figure.



To derive the formula for the surface area of such a solid, we adopt the same strategy as before: divide, approximate, sum, and take limit. We skip the derivation here (refer to the textbook if you are interested to see the derivation). We instead give the intuitive idea - **a self-study**.

. . . . . .

The idea is that the surface area of the solid in the figure can be obtained by summing up the surface areas of small pieces of the solid. Each small piece is called a frustum as shown in the figure below.



The surface area of a typical frustum is given by:

$$\textit{Circumference} \times \textit{height of frustum} = 2\pi f(x_i^*) \times \sqrt{1 + \left(f'(x_i^*)\right)^2} \cdot \Delta x.$$

Here the height of the frustum is the arc length of the straight line segment at the side of the frustum.

So the surface area of the whole solid of revolution is approximated by summing up the surface areas of all these approximating frustums:

$$\sum_{i=1}^{n} 2\pi f(x_i^*)\sqrt{1+\left(f'(x_i^*)\right)^2}\cdot\Delta x.$$

To get the exact surface area of the whole solid we take limits. Then the summation symbol $\sum$ *becomes the integration symbol* $\int_a^b$ *while the* Δx *symbol becomes the differential symbol* dx*:*

$$\int_a^b 2\pi f(x)\sqrt{1+\left(f'(x)\right)^2}\,dx.$$

Theorem (Surface Area)

If the function $f(x) \geq 0$ has a continuous first derivative (i.e. $f'(x)$ is continuous) on the interval $[a, b]$, then the surface area generated by rotating the curve $y = f(x)$ about the x-axis is given by

$$S = \int_a^b 2\pi f(x)\sqrt{1 + (f'(x))^2}\, dx.$$

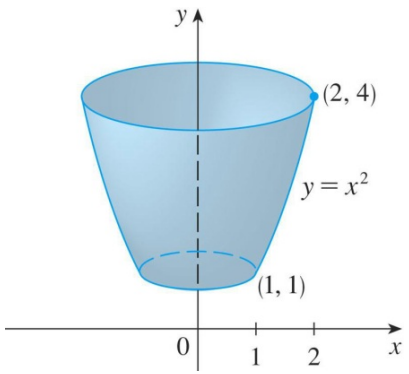
This is sometimes written as $\displaystyle\int_a^b 2\pi y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$.

We can also remember the formula in the following way: $2\pi f(x)$ is the circumference and $\sqrt{1 + (f'(x))^2}\, dx$ is the arc length.

Example

Find the surface area generated by rotating the parabola $y = x^2$ from $(1, 1)$ to $(2, 4)$ about the y-axis.

Solution. The solid is sketched in the figure below.



We need to apply the formula, except with the role of x and y reversed.

So we have

$$x = \sqrt{y} \quad \Longrightarrow \quad x' = \frac{1}{2\sqrt{y}}.$$

The formula

$$S = \int_a^b 2\pi f(x)\sqrt{1 + (f'(x))^2}\, dx$$

now becomes

$$S = \int_c^d 2\pi g(y)\sqrt{1 + (x')^2}\, dy,$$

where $[c, d]$ is the range of y, and $g(y)$ is the radius of each frustum.

We thus get

$$
\begin{aligned}
S &= \int_1^4 2\pi\sqrt{y}\cdot\sqrt{1+\left(\frac{1}{2\sqrt{y}}\right)^2}\,dy \\
&= \int_1^4 2\pi\sqrt{y}\cdot\sqrt{1+\frac{1}{4y}}\,dy \\
&= \pi\int_1^4 \sqrt{4y+1}\,dy = \pi\int_1^4 (4y+1)^{\frac{1}{2}}\,dy \\
&= \pi\left[\frac{(4y+1)^{\frac{3}{2}}}{\frac{3}{2}\cdot 4}\right]_1^4 = \frac{\pi}{6}\left(17^{\frac{3}{2}}-5^{\frac{3}{2}}\right) \\
&= \frac{\pi}{6}\left(17\sqrt{17}-5\sqrt{5}\right).
\end{aligned}
$$

END OF CHAPTER 4

Appendix to Chapter 4

In this Appendix, the following explanations are given:

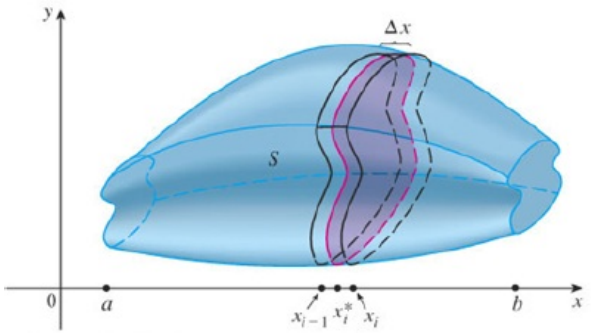
- Cross-sectional method I: General solids
- An alternative method to find volumes using cylindrical shells.

- **Cross-sectional method I: General solids**

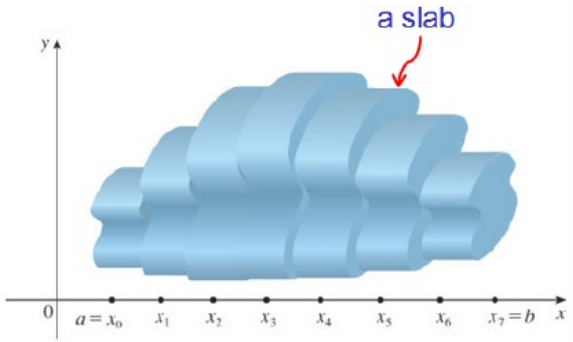
This method can be applied to any given solid, as long as the cross-sectional area $A(x)$ can be easily found.

The same idea as we previously did for the area problem is applied.

- The first step is to divide the given solid S into n “slabs" of equal width Δx. Each slab is divided by using the planes $P_{x_1}, P_{x_2}, \cdots, P_{x_n}$. To do this, we need to first divide the interval $[a, b]$ into n equal subintervals $[a, x_1], [x_1, x_2], \cdots, [x_{n-1}, b]$.



- Let’s zoom in on the ith interval $[x_{i-1}, x_i]$. The ith slab looks as follows.



- This portion of the solid S is not necessarily a cylinder, and so it is still not easy for us to calculate the area of this slab easily.

- Instead we approximate this slab with a cylinder. To do so we pick a sample point x_i^* from the ith interval $[x_{i-1}, x_i]$. This sample point can again be any point in the ith interval, including the left endpoint, right endpoint, or the midpoint of the interval.

- Using the sample point x_i^* we look at the cylinder with constant cross-section area $A(x_i^*)$ and cylinder height Δx. This cylinder has roughly the same volume as the volume of the ith slab. The volume of this cylinder is

$$\text{Cross-sectional area} \times \text{Height} = A(x_i^*) \cdot \Delta x.$$

- So it is reasonable to approximate the volume of the entire solid S by

$$V \approx \sum_{i=1}^{n} A(x_i^*) \cdot \Delta x.$$

- To turn this into the exact volume, we proceed as before. We take limit on the sum of volumes of approximating cylinders by letting $n \to \infty$. At the end, we get

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*) \cdot \Delta x.$$

- Comparing this with the definition of the definite integral, we can recognize the expression $A(x_i^*) \cdot \Delta x$ as the nth Riemann sum of the function $A(x)$. So, we have

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*) \cdot \Delta x = \int_a^b A(x)\, dx.$$

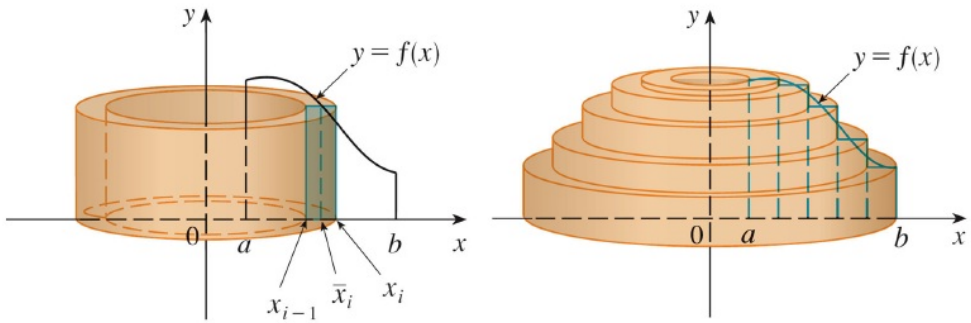
- **An alternative method to find volumes using cylindrical shells**

The same method as before is applied.

- First divide $[a, b]$ into n subintervals $[a, x_1], [x_1, x_2], \cdots, [x_{n-1}, b]$.

- Then divide the region under the curve $y = f(x)$ and approximate it by n strips.

- By rotating each strip about the y-axis we obtain a small circular piece of the solid. Putting each small circular piece together gives an approximation to the solid S.



As we have observed above the volume of the ith cylindrical shell is given by

$$(\text{Average circumference}) \times (\text{height}) \times (\text{thickness}) = (2\pi\bar{x}_i) \cdot f(\bar{x}_i) \cdot \Delta x.$$

Summing up the volumes of all the cylindrical shells gives

$$V \approx \sum_{i=1}^{n} (2\pi\bar{x}_i) \cdot f(\bar{x}_i) \cdot \Delta x.$$

Finally, taking limit as $n \to \infty$ to get the actual volume:

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} (2\pi\bar{x}_i) \cdot f(\bar{x}_i) \cdot \Delta x = \int_a^b 2\pi x f(x)\, dx.$$